Are the neurons of a reward-trained Ms. Pac-Man agent already clean? A preregistered neuron audit with causal steering

Libor "Burny" Burian

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Abstract

Goodfire's essay "Neural Networks Are Already Clean" argues that the individual neurons of trained vision and language networks are largely interpretable; in its preliminary Qwen3-32B audits, about 90% of neuron poles have "one main job". We test this claim, and whether single units can steer behaviour, in three deep RL agents that learned Atari Ms. Pac-Man from game reward alone. They are public Sample Factory APPO agents with a Nature-CNN torso, trained for 2B emulator frames (about 500M agent steps).

We re-implement the agents in PyTorch and check them against the original stack: observations are bit-exact and logits agree to 1.1e-05 on CPU (to 4.5e-02 on the GPU used for data collection). We decode exact game state from the console RAM, including frightened-mode registers that we identified independently from sprite colours (97.1% per-ghost agreement; already documented in jev_deep_rl), and label every unit against 40 ground-truth features under a preregistered protocol.

Not "already clean". On held-out games, the fraction of alive fc neurons with one main job is 0.8%, vs 0.2% for random directions and 9.4% for an untrained network. The preregistered privileged-basis test is supported for agent 1111, but the effect is trivially small: 3 of 367 neurons vs 1 of 512 random directions. It is not supported for agent 2222 and not supported for agent 3333. The agent is significantly *less* clean than non-negative unit mixtures in agents 1111 and 2222, and than the untrained network in all three. Exploratorily, every one-job unit, in every set, follows a single slow global variable (pellets left or the maze id). A lookup table over Ms. Pac-Man's maze position × heading explains a median out-of-sample R^2 of 0.20 per neuron, vs 0.03 for the best single feature, which suggests conjunctive, place-like codes.

One unit steers, but not locally. Two steering candidates were frozen before any held-out data. Setting a single conv3 unit (channel 18 at one spatial position) to its 99th percentile lowers the preregistered approach rate toward that position: 0.424 → 0.361, Holm p 0.0002. The effect is stronger than that of each of the 63 other channels at that position, each set to its own 99th percentile, so the preregistered H2 passes. Post-hoc checks on the same games limit what this shows:

- approach toward almost every maze location falls (the steered cell ranks 28 of 49);
- the agent vacates the bottom-left, not the steered bottom-right;
- the unit is the strongest perturbation at its cell (open-loop KL rank 1 of 64).

It is a causal handle on navigation, not a local "no-go zone".

Ablating fc neuron 453, which votes UP, reduces moving up toward edible ghosts. The reduction is specific to that context, and no null ablation matches it. It does not significantly reduce ghosts eaten per power pill, so H2 fails for this candidate. Its best direction label ("edible ghost is up") does not survive confound control.

Replication. Frozen selection rules applied to the other two agents give the following (H3):

- most-repelling-patch rule: replicates as preregistered. Post hoc, it always selects the strongest patch at its cell. The agent clearly leaves the steered region only in agent 3333 (partly in 2222);
- ghost-hunting ablation rule: does not replicate (exploratory, because fc 453 failed H2).

All tests, deviations, errata and negative results are reported. That includes an independent audit whose post-hoc checks (Section 4.3, `reports/00_summary.md` errata 11–16) correct an earlier "no-go zone" reading of the conv3 result.

1. Introduction

Motivation. Mechanistic interpretability asks whether the internal units of a network can be read as human-understandable variables and whether manipulating them changes behaviour in the predicted way.

The claim we test. Goodfire recently argued that the neurons of trained networks are already largely interpretable (Cammarata, Goodfire 2026). In its preliminary audits of Qwen3-32B, "roughly 90% of neuron poles have one main job"; it makes the same argument for vision models such as InceptionV1 and ConvNeXt. Its evidence combines top-activating examples, feature visualisation, steering and attribution graphs.

Why RL is a sharp test. Reinforcement-learning agents trained from reward alone offer two advantages:

- the environment state is fully known (for Atari, the console RAM);
- behaviour can be measured causally in closed loop.

Gap in prior work. Prior interpretability work on Ms. Pac-Man used:

- input saliency (Greydanus et al. 2018);
- attention maps (Mott et al. 2019);
- NMF concepts (Bekkemoen & Langseth 2025);
- a sparse autoencoder without interventions (DuPlessie 2024).

Single-channel steering of RL agents has been shown for a Procgen maze agent (Mini et al. 2023). In a Sokoban planning agent, plans have been steered with probe directions (Taufeeque et al. 2024; Bush et al. 2025) and with channel-level interventions (Taufeeque et al. 2025). To our knowledge, no published work has steered individual neurons of a reward-trained Ms. Pac-Man agent.

Contributions.

1. A dependency-light re-implementation of the public Sample Factory Ms. Pac-Man agents with a golden fidelity test (bit-exact observations; logits agree to 1.1e-5 on CPU).
2. RAM ground truth for Ms. Pac-Man, including a pixel validation of the frightened-mode register `ram[116]` and the per-ghost edible bits (bit 7 of `ram[1+i]`), and a maze lattice graph. We found these registers independently; jev_deep_rl documented them first.
3. A preregistered Goodfire-style audit of every alive fc neuron against three controls (H1).
4. Preregistered causal steering tests with matched null units on held-out games (H2), with replication rules for two more agents (H3).

2. Background and related work

- **Neuron-level interpretability.**
  - Circuits (Olah et al. 2020).
  - Polysemanticity and superposition, which motivate dictionary learning (Bricken et al. 2023).
  - Goodfire's counter-argument that neurons are mostly clean (Cammarata, Goodfire 2026).
  - Direct logit attribution (Elhage et al. 2021) and occlusion maps (Zeiler & Fergus 2014).
- **RL interpretability.**
  - An Atari model zoo with activation analysis (Such et al. 2019).
  - CoinRun attribution and model editing (Hilton et al. 2020).
  - Probe-based steering of planning in Sokoban (Bush et al. 2025).
  - Attention maps of Ms. Pac-Man agents (Mott et al. 2019).
- **Ground truth for Atari.**
  - AtariARI RAM annotations (Anand et al. 2019).
  - OCAtari object extraction (Delfosse et al. 2023).
  - Ms. Pac-Man frightened-mode registers (jev_deep_rl, 2026).

- **Evaluation protocol.** Sticky actions (Machado et al. 2018).

3. Method

3.1 Agents and fidelity

- **Agents.** `edbeeching/atari_2B_atari_mspacman_{1111,2222,3333}`:
  - Sample Factory 2.0 APPO (Petrenko et al. 2020), an asynchronous variant of PPO (Schulman et al. 2017); 2B emulator frames (about 500M agent steps at frameskip 4), clipped reward, $\gamma = 0.99$, no RNN;
  - Nature-CNN torso (Mnih et al. 2015) (conv 32/64/64; Sample Factory's `convnet_atari`) → fc 512 ReLU → linear policy (9 actions) and value heads.
- **Environment.** The wrapper chain is `MsPacmanNoFrameskip-v4` → no-op reset → max-and-skip 4 → episodic life → reward clipping → 84×84 grayscale → stack 4.
- **Fidelity.** We reproduce the wrapper chain and network in PyTorch and verify both against a golden trajectory generated with sample-factory 2.0.3, gym 0.26.2 and ale-py 0.8.1:
  - 0 mismatched observations in 446 steps;
  - max logit difference 1.1e-05 on CPU. On the GPU used for all data collection (PyTorch's default cuDNN TF32 convolutions) it is 4.5e-02. The test covers CPU only (Section 6).

  Appendix A has the details.

3.2 Ground truth

- **RAM decoding** follows OCAtari and AtariARI.
- **Frightened mode** was identified by correlating RAM bytes with blue ghost pixels and power-pill events. The same registers were documented earlier in jev_deep_rl, which also notes that only the low six bits of `ram[116]` count down.
- **Maze.** A 14×18 lattice graph built from the frame gives breadth-first-search distances and directions.
- **Features.** 40 features in five families (Appendix B). A step is excluded when Ms. Pac-Man's RAM position is unchanged from t−1 to t and from t to t+1. This removes death animations, ready screens and level transitions, and also standstills such as pushing into a wall.

3.3 Data and stochasticity

- **Splits.** 100 discovery games (seeds 0–99) and 100 held-out games (seeds 10000–10099) per agent.
- **Action sampling.** Actions are sampled from the policy with a per-game RNG, so steered and baseline games share random numbers.
- **Stochasticity fallback.** A preregistered check found that without sticky actions, same-start games stayed identical for a median of 1230 steps. The preregistered fallback (sticky actions, $p = 0.25$) was therefore used for all data (median divergence 320 steps).

3.4 Audit (H1)

- **Per-unit score.** For each alive fc unit (active on ≥ 0.1% of in-control discovery steps; positive pole = post-ReLU activation):
  - R^2_1, the out-of-sample R^2 of bin means of each single feature (20 quantile bins; features with ≤ 20 distinct values use their values);
  - R^2_all, the out-of-sample R^2 of histogram gradient boosting (scikit-learn; LightGBM-style, Ke et al. 2017) on all features.
- **One-job criterion.** R^2_all ≥ 0.3 and max $R^2_1 \geq 0.7 \cdot R^2$_all.
- **Controls.**
  - (a) random orthonormal directions;
  - (b) non-negative mixtures of 2–8 units;
  - (c) an untrained, orthogonally initialised network (Saxe et al. 2014) on the same states (its alive units; init seed 0 for every agent).

- **Test.** Game-cluster bootstrap (Field & Welsh 2007) of the difference in one-job fraction. The fitted models are fixed; only the scored games are resampled. H1 is supported if the 95% CI against (a) excludes 0.

3.5 Steering (H2) and replication (H3)

- **Why intervene upstream of fc.** Because the heads are linear and there is no recurrence:
  - clamping fc unit n to v replaces its state-dependent vote $h_n(s) \cdot W_\pi[:, n]$ with the constant $v \cdot W_\pi[:, n]$, so it is a constant action bias wherever the unit is inactive;
  - ablating it removes the state-dependent vote;
  - conv3 interventions act through 512 ReLUs.
- **Candidate selection** (discovery data only):
  1. Exact open-loop screens, using a rank-1 update for single conv3 units.
  2. Closed-loop pilots on discovery seeds.
  3. Two candidates frozen in a committed addendum.
- **Confirmatory design** (held-out seeds):
  - 100 baseline games, 50 games per candidate, and 20 games per null unit;
  - nulls: 99 random alive fc units, or the 63 other conv3 channels at the same cell, each at its own p99;
  - pass criteria: predicted sign with a Holm-corrected (Holm 1979) paired sign-flip $p < 0.05$ (10,000 permutations, so $p \geq 1/10{,}001$; Phipson & Smyth 2010), **and** an effect beyond the 95th percentile of the null;
  - off-target metrics: two-sided tests, Benjamini–Hochberg-corrected (Benjamini & Hochberg 1995);
  - approach rate: the share of in-control steps after which the BFS distance to the target node is smaller 8 agent steps later.
- **Replication (H3).** Deterministic selection rules transfer the procedure to agents 2222 and 3333. A rule replicates if its unit passes both criteria in both agents, after a further Holm correction over the two agents. The preregistration defines H3 only for candidates that pass H2. fc 453 failed, so its rule's replication is exploratory.
- **Pairing check (exploratory).** An unsteered A/A arm replays 50 held-out seeds in the steering runner. In agent 1111, 50 of 50 games match their paired baseline action for action (2222: 50 of 50; 3333: 50 of 50). So GPU batching does not break the seed pairing.

4. Results

4.1 Fidelity and ground truth

check	value
golden trajectory steps (incl. 2 life losses and game over)	446
steps with any observation pixel mismatch	0
max abs. logit difference (CPU)	1.1e-05
max abs. value difference (CPU)	4.8e-07
max abs. logit difference on GPU (NVIDIA GeForce RTX 5070 Laptop GPU, cuDNN TF32 on, as in data collection)	4.5e-02
reference stack	sample-factory 2.0.3, gym 0.26.2, ale-py 0.8.1, torch 1.13.1 (WSL)

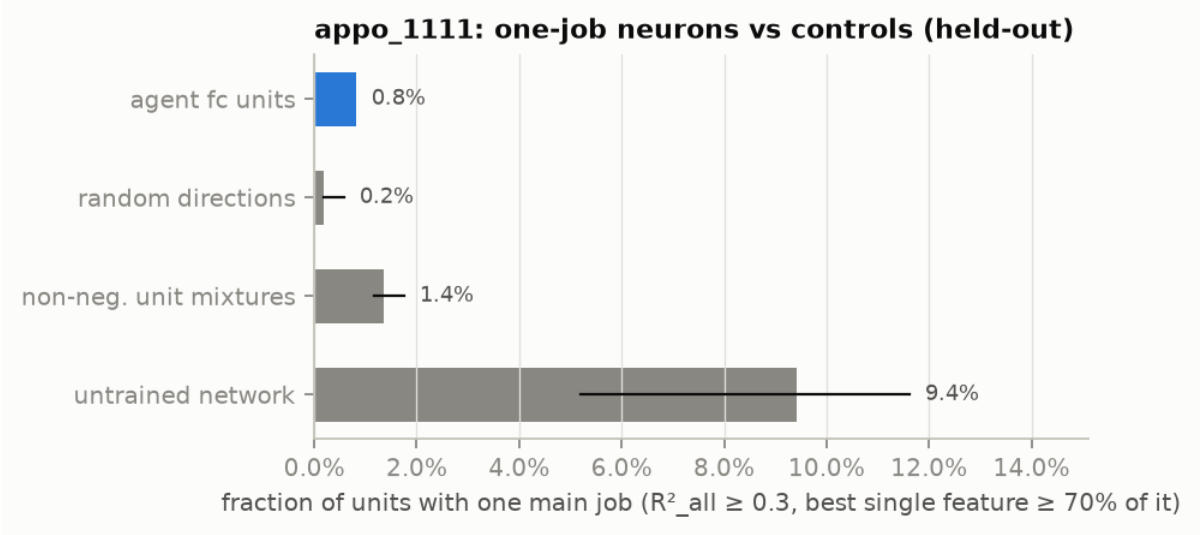
validation (discovery games, sticky actions)	value
agent steps checked	161995 (100 games)
RAM edible bit (bit 7 of ram[1+i]) vs blue sprite pixels, per ghost-step	97.1%
frightened timer ram[116] > 0 vs any ghost edible	100.0%
eaten pellets at Ms. Pac-Man's lattice node (26825 events)	100.0%

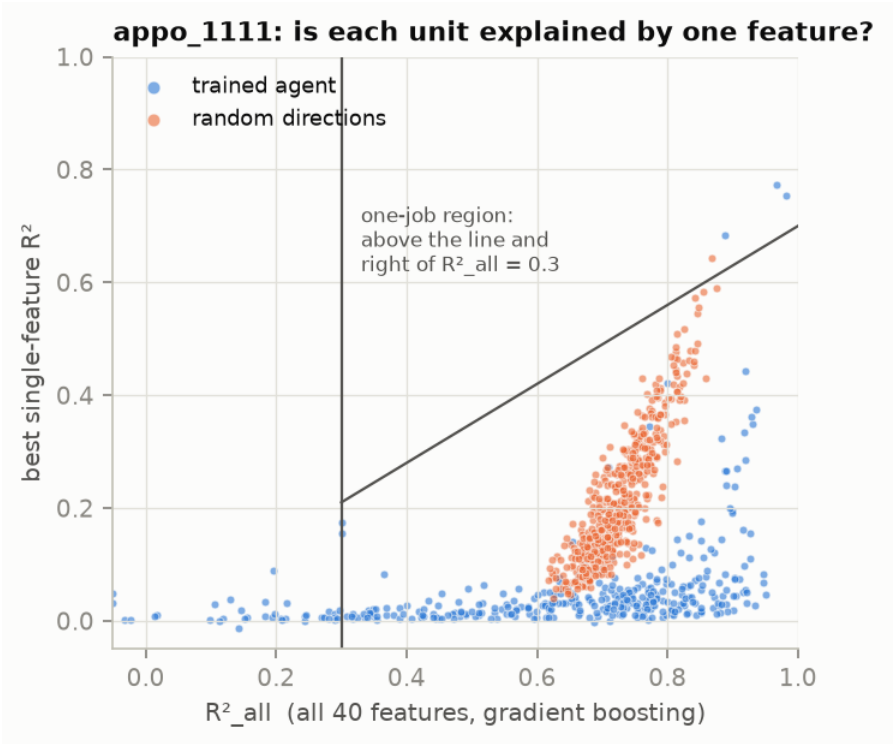
validation (discovery games, sticky actions)	value
steps in control (position changed between t−1 and t or t and t+1; excludes death, ready, level transitions and standstills)	85.1%
distinct maze layouts seen	4
edible-bit vs blue-pixel disagreements (18585 ghost-steps): bit set but no blue pixels	95.5%
… median countdown (low 6 bits of ram[116]) at a disagreement / share at ≤ 10 (all edible ghost-steps: 7.1%)	13 / 38.2%
countdown start value when a pill is eaten in normal mode (207 pills)	38–62
countdown drop per 4 agent steps (modal value, share)	2 (97.5%)
frightened steps with the upper bits of ram[116] set	18.5%

4.2 H1: one-job neurons

unit set (held-out, confirmatory)	units	one-job fraction [95% CI]	median R^2_all	median best single R^2
trained agent (fc units)	367	0.8% [0.8%, 0.8%]	0.708	0.031
random directions (control a)	512	0.2% [0.2%, 0.6%]	0.719	0.185
non-negative unit mixtures (control b)	512	1.4% [1.2%, 1.8%]	0.720	0.053
untrained network (control c)	499	9.4% [5.2%, 11.6%]	0.537	0.096

H1 test (agent minus random directions): 0.6% [0.2%, 0.6%]; H1 supported. One-job units: agent 3 of 367, random directions 1 of 512. Secondary controls (not part of the verdict): agent minus non-negative mixtures -0.5% [-0.9%, -0.4%]; agent minus untrained network -8.6% [-10.8%, -4.4%].





Robustness. The table below adds a start-state-cluster bootstrap that resamples groups of games sharing their first in-control RAM state. There are only 4 such groups among the held-out games, so these CIs are coarse. The threshold sweep shows that agent 1111's small advantage over random directions reverses at a looser ratio threshold: at a ratio of 0.5, random directions score 8.0% vs the agent's 1.4%.

What the one-job units are (exploratory, discovery split-half). Every one-job unit follows one slow global variable:

- agent: n_pellets for 3 of 3;
- random directions: n_pellets for 3 of 3;
- non-negative mixtures: n_pellets for 8 of 8;
- untrained network: maze for 53 of 53.

With this dictionary the criterion mostly detects units that follow pellets left or the maze colour, so the H1 verdict in either direction says little about semantic interpretability.

unit set	one-job fraction	game-bootstrap 95% CI	start-state-cluster 95% CI
trained agent (fc units)	0.8%	[0.8%, 0.8%]	[0.8%, 0.8%]
random directions (control a)	0.2%	[0.2%, 0.6%]	[0.2%, 0.4%]
non-negative unit mixtures (control b)	1.4%	[1.2%, 1.8%]	[1.4%, 2.0%]
untrained network (control c)	9.4%	[5.2%, 11.6%]	[4.8%, 11.0%]

Threshold sweep, one-job fraction shown as agent / random directions (rows: R^2_all threshold; columns: best-single / R^2_all ratio threshold). Start-state clusters: 4.

R^2_all ≥	ratio ≥ 0.5	0.6	0.7	0.8	0.9
0.1	1.6% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%
0.2	1.6% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%
0.3	1.4% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%

R^2_all ≥	ratio ≥ 0.5	0.6	0.7	0.8	0.9
0.4	1.1% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%
0.5	1.1% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%

Exploratory conjunctions (not preregistered; discovery split-half, fitted on even games and scored on odd ones). Bin counts differ: place has up to 252 lattice nodes and place × heading up to 1,008 bins, while single features have ≤ 20 bins. Gradient boosting on all 40 features reaches a median of 0.72 on the same split:

explanatory variable(s) (exploratory, post hoc)	median R^2	units with $R^2 > 0.3$	units with $R^2 > 0.5$
place (lattice node)	0.127	11.7%	2.5%
place x heading	0.202	30.5%	7.4%
place x frightened	0.165	23.4%	6.0%
pellets left (20 bins) x frightened	0.022	2.7%	1.1%
danger distance (20 bins) x danger direction	0.018	0.3%	0.0%

4.3 H2: steering

intervention	primary metric (predicted sign)	baseline → steered	Δ [95% CI]	Holm p	above null 95th pct?	H2
fc neuron 453 zero-ablated	ghosts_per_pill (−)	0.294 → 0.228	-0.066 [-0.192, 0.101]	0.2280	yes (null q95 0.013)	fail
conv3 channel 18 set to its p99 at cell (5,6)	approach (−)	0.424 → 0.361	-0.063 [-0.077, -0.050]	0.0002	yes (null q95 0.013)	pass
fc neuron 188 clamped to its p99 (constant logit bias, DLA UP -0.70) (positive control)	action_given_trigger (−)	0.395 → 0.305	-0.090 [-0.102, -0.079]	–	–	–

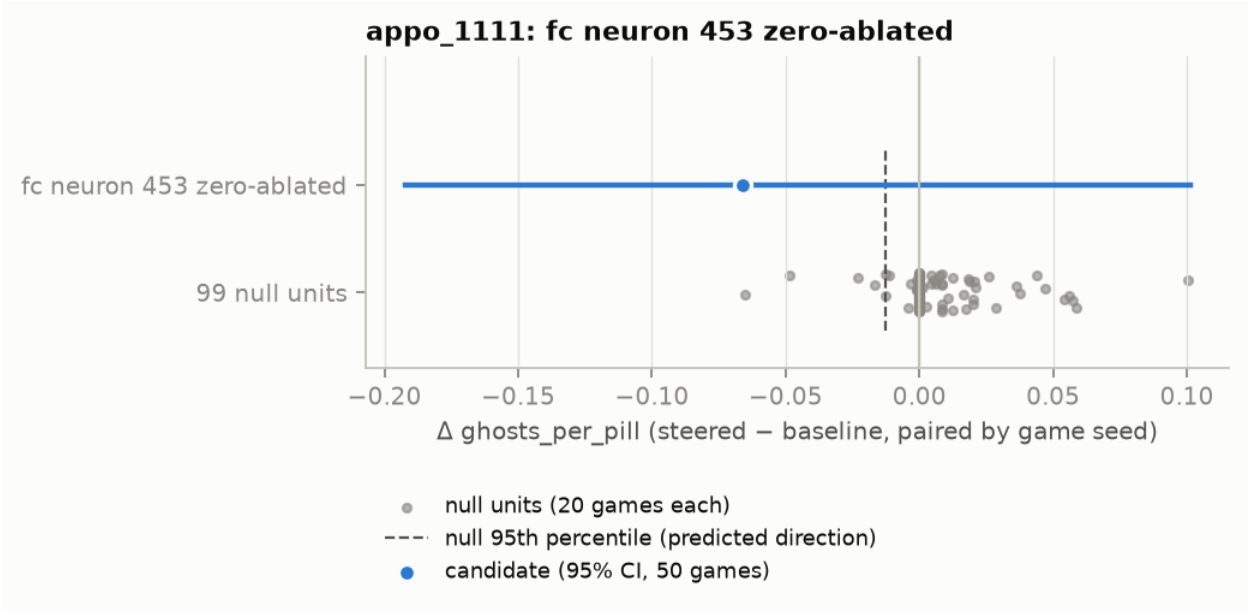
p = 0.0001 is the smallest attainable value of the 10,000-permutation sign-flip test (1/10,001); 0.0002 is that floor after Holm over 2 tests, so it means p ≤ that value. The positive control's frozen label says "constant logit bias": clamping unit n to v shifts the logits by $(v - h_n(s)) \cdot W_\pi[:, n]$, which is constant only on steps where the unit is inactive (fc 188: active on 2.1% of in-control discovery steps).

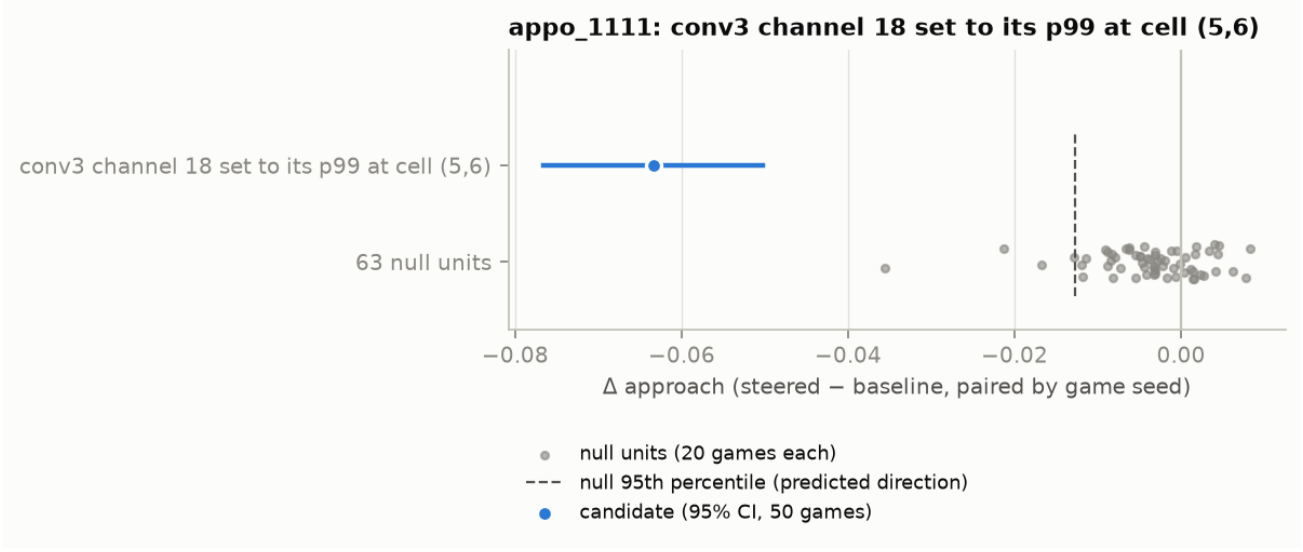
Off-target effects (two-sided paired sign-flip tests, Benjamini-Hochberg-corrected):

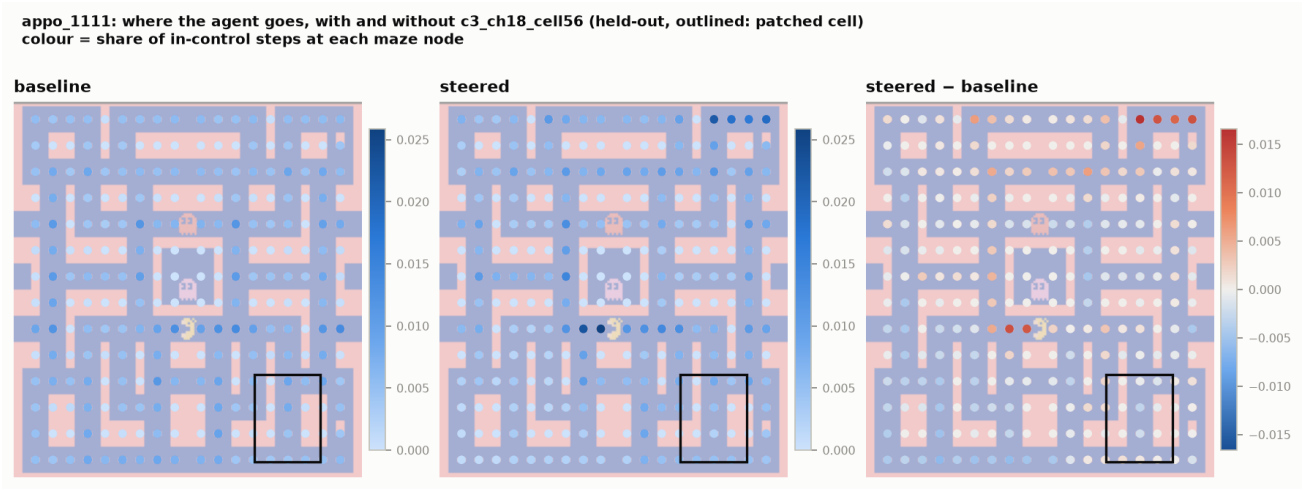
intervention	off-target metric	Δ [95% CI]	two-sided p	BH q
fc neuron 453 zero-ablated	return	-1619 [-2194, -1056]	0.0001	0.0001

intervention	off-target metric	Δ [95% CI]	two-sided p	BH q
fc neuron 453 zero-ablated	deaths_per_1k	1.14 [0.75, 1.52]	0.0001	0.0001
fc neuron 453 zero-ablated	pellets_per_1k	-9.23 [-17.68, -0.84]	0.0416	0.0468
fc neuron 453 zero-ablated	in_control	-469 [-633, -304]	0.0001	0.0001
conv3 channel 18 set to its p99 at cell (5,6)	return	-2490 [-2990, -2005]	0.0001	0.0001
conv3 channel 18 set to its p99 at cell (5,6)	deaths_per_1k	4.17 [3.27, 5.14]	0.0001	0.0001
conv3 channel 18 set to its p99 at cell (5,6)	pellets_per_1k	-12.16 [-20.78, -3.83]	0.0071	0.0091
conv3 channel 18 set to its p99 at cell (5,6)	ghosts_per_pill	-0.05 [-0.18, 0.09]	0.5156	0.5156
conv3 channel 18 set to its p99 at cell (5,6)	in_control	-788 [-944, -633]	0.0001	0.0001

p = 0.0001 is the smallest attainable value of the 10,000-permutation sign-flip test (1/10,001), so it means p ≤ that value.







Post-hoc specificity checks (added by an independent audit after all confirmatory results; same held-out games, no new games; `specificity` phase).

The preregistered approach metric only scores the steered cell's target node. We therefore computed it for the target nodes of all 49 conv3 cells:

- **The conv3 drop is not local.** The steered cell ranks 28 of 49 (1 = largest drop), and the median cell's Δ is -0.068.
- **The agent leaves a different region.** It vacates the bottom-left quadrant (0.264 → 0.182 of in-control time). The steered bottom-right barely changes (0.268 → 0.254).
- **The null is weaker.** Channel 18's open-loop KL at the cell ranks 1 of 64 (10× the median of the other 63). Its return loss exceeds 100.0% of its 63 null units. The preregistered null matched the dose rule, not the perturbation strength.
- **Navigation degrades.** The action-change rate goes 0.483 → 0.565, and the 8-step net displacement goes 2.23 → 1.97 lattice steps. Windows with a life loss do not explain the drop (Δ without them -0.065).

The fc 453 ablation, by contrast, is context-specific. P(UP) changes by Δ -0.190 [-0.209, -0.171] when the nearest edible ghost is up, Δ -0.028 [-0.044, -0.013] in other frightened states with up open, and Δ 0.030 [0.012, 0.049] in normal mode. 0 of 99 null ablations reduce the secondary metric as much (seeds 10000–10019).

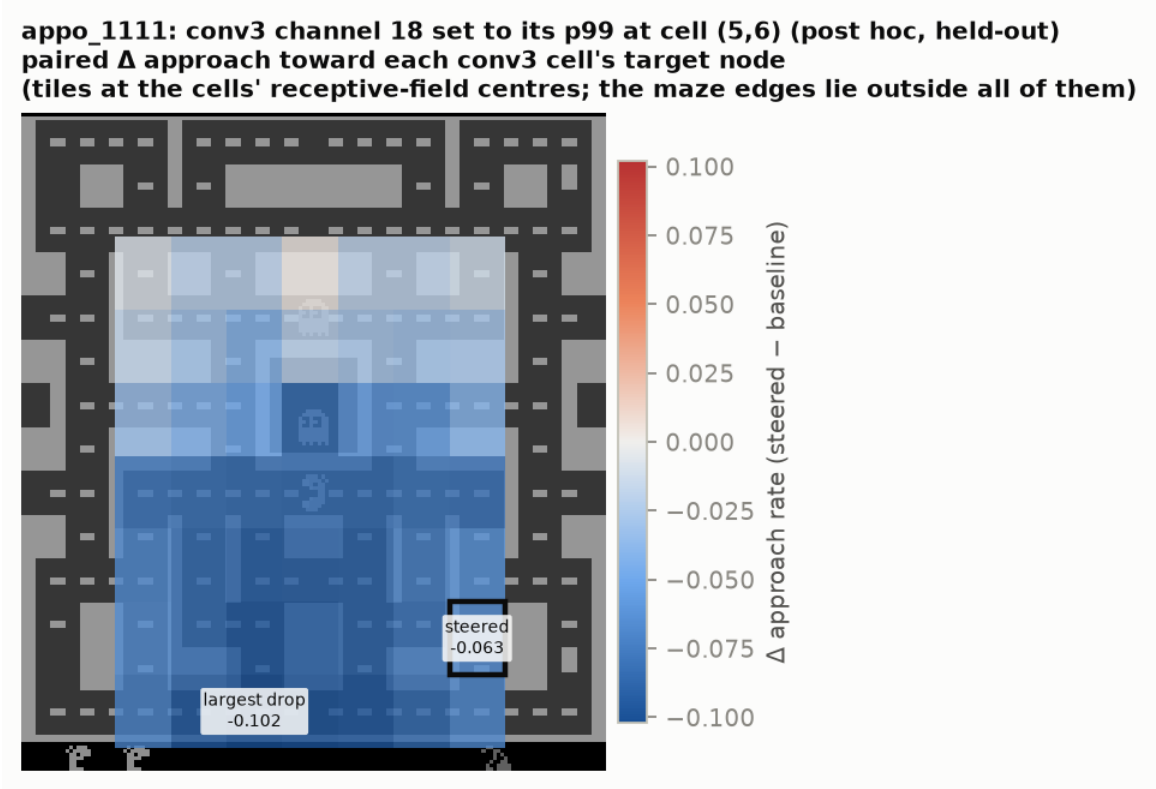
conv3 candidate (post hoc)	Δ approach, steered cell (rank of 49, 1 = largest drop)	median Δ over all 49 cells	time within 3 steps of target (rank of 49)	occupancy of the steered cell's quadrant	most vacated quadrant
conv3 channel 18 set to its p99 at cell (5,6)	-0.063 (28)	-0.068	0.062 → 0.057, -9% (20)	bottom-right: 0.268 → 0.254 (Δ -0.014 [-0.037, 0.010])	bottom-left: 0.264 → 0.182 (Δ -0.082 [-0.110, -0.056])

conv3 candidate (post hoc)	open-loop KL at its cell: rank of 64 (× median of the other 63)	nulls with a larger return loss (seeds 10000–10019)	action-change rate	8-step net displacement (lattice steps)	Δ approach without life-loss windows
conv3 channel 18 set to its p99 at cell (5,6)	1 (10×)	0.0% of 63	0.483 → 0.565 (Δ 0.082 [0.060, 0.104])	2.23 → 1.97 (Δ -0.26 [-0.30, -0.22])	-0.065

fc candidate (post hoc)	P(voted direction) when the nearest edible ghost is in that direction	frightened, edible ghost elsewhere, direction open	normal mode, direction open	secondary Δ vs 99 null ablations (seeds 10000–10019): nulls at least as strong / null q95 (signed, as in the steering table)
fc neuron 453 zero-ablated (up)	0.505 → 0.315 (Δ -0.190 [-0.209, -0.171])	0.430 → 0.402 (Δ -0.028 [-0.044, -0.013])	0.342 → 0.372 (Δ 0.030 [0.012, 0.049])	Δ -0.194: 0 of 99 / 0.004

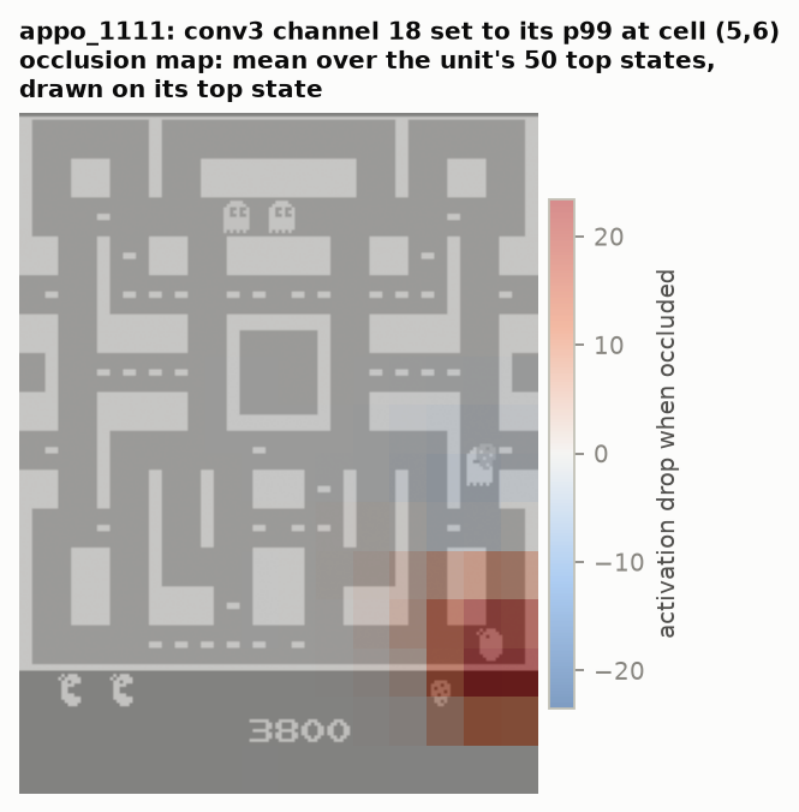
Between-game dependence: the held-out games fall into 4 start-state clusters (sizes 24, 24, 25, 27).

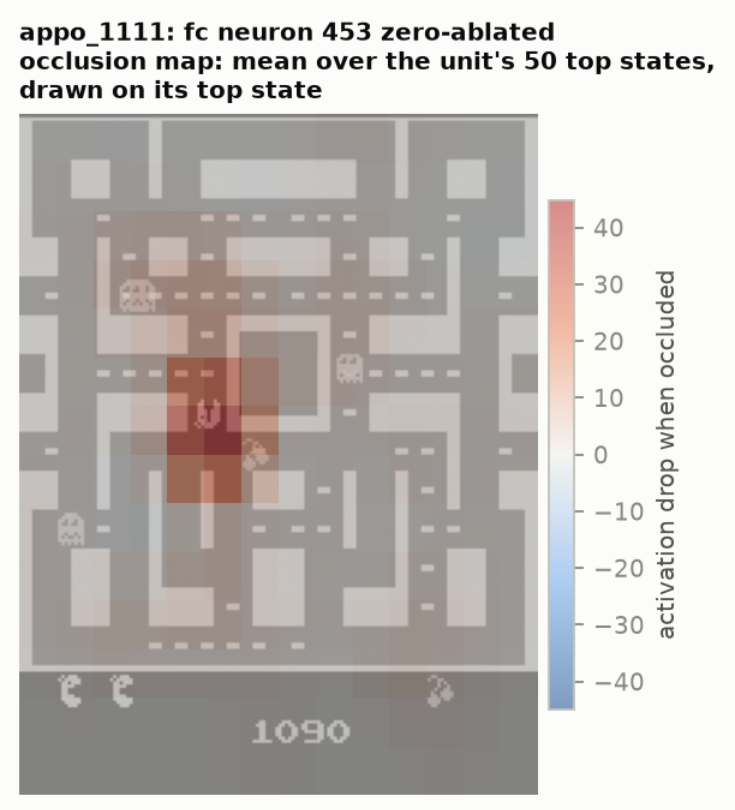
candidate	ICC of paired Δ within clusters	design effect	z	cluster-adjusted z	cluster-adjusted Holm p
fc neuron 453 zero-ablated	0.072	1.83	-0.88	-0.65	0.2579
conv3 channel 18 set to its p99 at cell (5,6)	0.125	2.44	-9.39	-6.00	< 0.0001



Attribution of the candidate units: exact upstream contributions, occlusion peak, and confound-controlled label checks (stratified AUROC within frightened × player-region strata; unique R^2 = drop-one loss of the gradient-boosting ceiling):

unit	strongest upstream units (mean exact contribution)	occlusion peak (screen x, y)	label checks: single-feature R^2 / stratified AUROC / unique R^2
fc neuron 453 zero-ablated	conv3 ch 17 (+5.1), conv3 ch 59 (+4.4), conv3 ch 1 (+4.3)	(57, 90)	dir_edible_up: 0.14 / 0.59 / 0.003; n_pellets: 0.38 / – / 0.007; n_edible: 0.29 / – / 0.002
conv3 channel 18 set to its p99 at cell (5,6)	conv2 ch 40 (+6.9), conv2 ch 12 (+5.5), conv2 ch 56 (+5.1)	(137, 165)	–





The conv3 unit's *activation* depends on input inside its own receptive field: Ms. Pac-Man is within 12 px of its occlusion peak in 98.0% of its 50 top states (exploratory). Its *steering effect*, as shown above, is not local.

fc 453's best direction label ("edible ghost is up") does not survive confound control. The unit is active mostly in

frightened mode (88.8% of its active steps vs 40.3% overall), and Ms. Pac-Man is near its occlusion peak in 80.0% of its 50 top states. Those top states largely repeat one early-game moment (86.0% of them within 10 steps of step 379, from 46 games), so its occlusion map describes one situation.

Its ablation acts through its UP vote, and the context control above shows where: the vote matters mainly when an edible ghost is up. This is post-hoc evidence, from one agent, that the unit's behavioural role is "chase the edible ghost upward", even though no single "ghost is up" feature predicts its activation well.

Discovery-phase evidence that led to the candidates (exploratory):

rank	conv3 channel @ cell	Δ P(move toward cell) at junctions	dose (p99)
repel 1	ch 18 @ (5, 6)	-5.8 pp	24.6
repel 2	ch 15 @ (0, 6)	-5.1 pp	27.2
repel 3	ch 40 @ (4, 6)	-4.0 pp	34.8
repel 4	ch 36 @ (5, 6)	-3.6 pp	22.7
repel 5	ch 58 @ (5, 6)	-3.6 pp	18.4
attract 1	ch 16 @ (6, 2)	+3.6 pp	23.0
attract 2	ch 8 @ (1, 0)	+3.0 pp	21.6
attract 3	ch 38 @ (5, 0)	+3.0 pp	20.0
attract 4	ch 34 @ (6, 0)	+2.7 pp	28.1
attract 5	ch 46 @ (6, 0)	+2.6 pp	14.5

intervention (discovery pilot, paired games)	targeted metric	baseline → steered	Δ [95% CI]	Δ return	Δ deaths / 1k
`c3_ch20_ablate`	deaths_per_1k	2.536 → 2.730	0.194 [-0.183, 0.671]	-210	0.19
`c3_ch20_clamp_frightened`	ghosts_per_pill	0.392 → 0.213	-0.179 [-0.292, -0.060]	-3668	5.28
`c3_ch20_clamp`	deaths_per_1k	2.536 → 9.071	6.535 [5.265, 7.759]	-4047	6.54
`c3_ch39_ablate`	deaths_per_1k	2.536 → 4.169	1.633 [0.646, 2.502]	-2566	1.63
`c3_ch53_ablate`	deaths_per_1k	2.536 → 2.665	0.129 [-0.035, 0.305]	-229	0.13
`c3_ch18_cell56`	approach	0.430 → 0.375	-0.056 [-0.068, -0.044]	-3520	5.57
`fc378_ablate`	action_given_trigger	0.502 → 0.475	-0.027 [-0.055, -0.000]	-1307	0.67
`fc453_ablate`	action_given_trigger	0.483 → 0.298	-0.184 [-0.204, -0.164]	-2739	1.52
`fc188_clamp`	action_given_trigger	0.384 → 0.302	-0.082 [-0.091, -0.072]	-3047	1.43

4.4 H3: replication

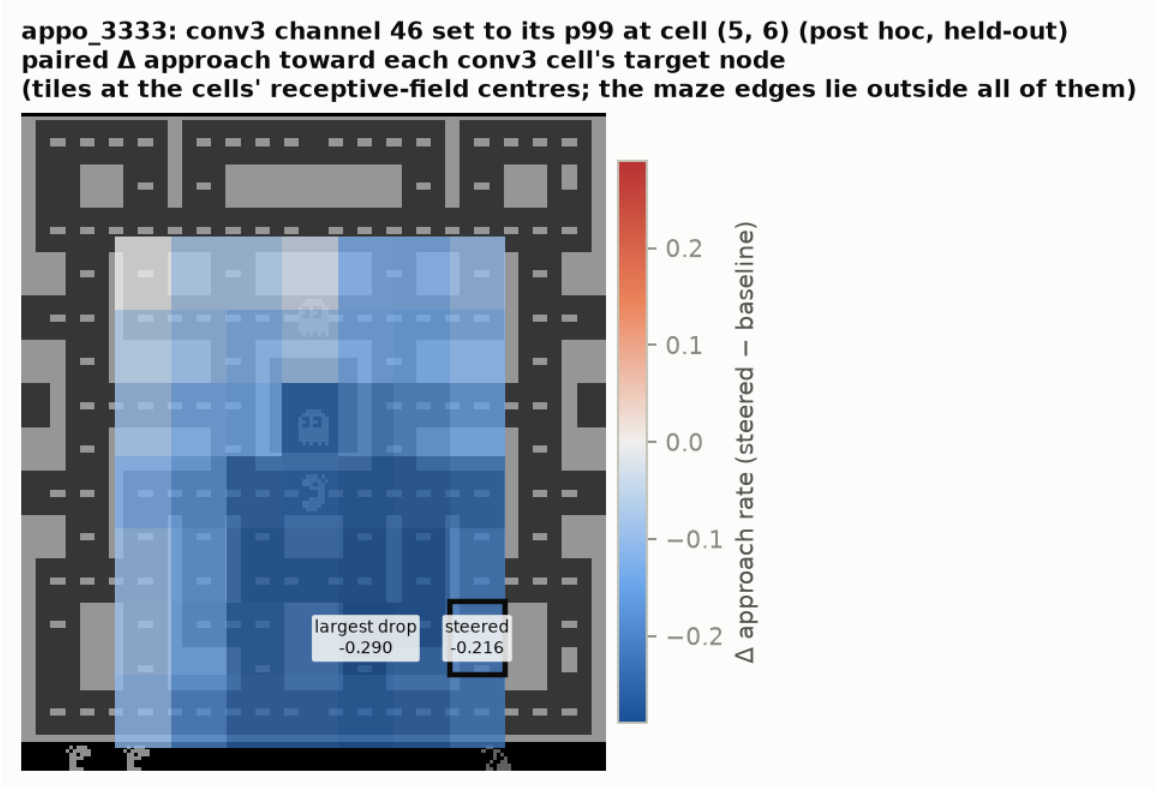
rule (frozen on agent 1111)	agent	selected unit	primary metric Δ [95% CI]	Holm p (within agent)	null units at least as strong	passes
zero-ablate the fc unit most needed for approaching edible ghosts	appo_2222	fc neuron 379 zero-ablated	-0.101 [-0.167, -0.036]	0.0025	0 of 99	yes
zero-ablate the fc unit most needed for approaching edible ghosts	appo_3333	fc neuron 351 zero-ablated	0.190 [0.071, 0.301]	0.9992	99 of 99	no
set the most repelling conv3 (channel, cell) to its p99	appo_2222	conv3 channel 3 set to its p99 at cell (0, 0)	-0.034 [-0.043, -0.026]	0.0002	0 of 63	yes
set the most repelling conv3 (channel, cell) to its p99	appo_3333	conv3 channel 46 set to its p99 at cell (5, 6)	-0.216 [-0.232, -0.200]	0.0002	0 of 63	yes

A rule replicates if its unit passes H2 criteria 1-2 in both agents and the within-agent Holm p-values stay < 0.05 after a further Holm correction over the two agents (preregistration §7, "Holm over seeds"). Preregistration §7 defines H3 only for candidates that passed H2; fc 453 failed H2 in agent 1111, so the fc-ablation rule's replication is exploratory. p = 0.0001 is the smallest attainable value of the 10,000-permutation sign-flip test (1/10,001); 0.0002 is that floor after Holm over 2 tests, so it means p ≤ that value. Post hoc (`specificity` phase): the conv3 rule picks the most repelling patch of the open-loop screen, and that patch's open-loop KL ranks 1 of 64 (appo_1111), 1 of 64 (appo_2222), 1 of 64 (appo_3333) among the channels at its cell, so the 63 null channels are all weaker open-loop perturbations; beating them does not show that the effect is specific to that channel's content.

Post hoc, the same specificity checks in the replication agents show a mixed picture:

- **Not local in either agent.** The steered cell's approach drop ranks 28 of 49 (2222) and 21 of 49 (3333).
- **Strongest perturbation in both.** The selected patch's open-loop KL ranks 1 of 64 and 1 of 64 at its cell.
- **Region avoided clearly in 3333, partly in 2222.** In agent 3333 the agent abandons the right third of the maze (0.336 → 0.049), which contains the steered cell. Agent 2222 leaves the whole top half (top third 0.348 → 0.259). Its most vacated quadrant is the top-right, while the steered cell is in the top-left.

Full tables are in `reports/05_replication.md`.



Steering per agent (held-out):

intervention	primary metric (predicted sign)	baseline → steered	Δ [95% CI]	Holm p	above null 95th pct?	H2
fc neuron 379 zero-ablated	ghosts_per_pill (−)	0.463 → 0.362	-0.101 [-0.167, -0.036]	0.0025	yes (null q95 0.027)	pass
conv3 channel 3 set to its p99 at cell (0, 0)	approach (−)	0.430 → 0.396	-0.034 [-0.043, -0.026]	0.0002	yes (null q95 0.007)	pass

p = 0.0001 is the smallest attainable value of the 10,000-permutation sign-flip test (1/10,001); 0.0002 is that floor after Holm over 2 tests, so it means p ≤ that value.

intervention	primary metric (predicted sign)	baseline → steered	Δ [95% CI]	Holm p	above null 95th pct?	H2
fc neuron 351 zero-ablated	ghosts_per_pill (−)	0.660 → 0.850	0.190 [0.071, 0.301]	0.9992	no (null q95 0.019)	fail
conv3 channel 46 set to its p99 at cell (5, 6)	approach (−)	0.440 → 0.224	-0.216 [-0.232, -0.200]	0.0002	yes (null q95 0.012)	pass

p = 0.0001 is the smallest attainable value of the 10,000-permutation sign-flip test (1/10,001); 0.0002 is that floor after Holm over 2 tests, so it means p ≤ that value.

5. Discussion

Are the neurons clean?

- **Not under single-feature labels.** With a 40-feature dictionary of single game variables, almost no fc neuron of these agents has one main job, and the agents are no cleaner than random directions in a meaningful way.
- **What the criterion detects.** Every one-job unit follows one slow global variable. For the untrained network that is the maze id (maze for 53 of 53), plausibly through the wall colour. For the agent it is pellets left (n_pellets for 3 of 3). This is why the untrained network scores highest, and why the agent is less clean than non-negative unit mixtures (agent minus mixtures -0.5% [-0.9%, -0.4%]). With this dictionary the criterion measures something closer to "follows a slow global variable" than to semantic selectivity.
- **The basis is still not arbitrary.**
  - Random directions score *higher* best-single-feature R^2 than neurons (Fig. R^2). Dense mixtures pick up slow global variables: pellets left is the best single feature for 472 of 512 random directions vs 155 of 367 neurons (discovery split-half).
  - Neurons are sparse (the median alive unit is active on 1.0% of in-control steps). Exploratorily, they are conjunctive: position × heading explains them far better than any single feature, and position (itself a joint x × y variable) does most of that.
- **What the units are.** They resemble place cells more than "ghost detectors". Greydanus et al. report something similar: several partially trained A3C Ms. Pac-Man agents "appeared to be tracking corridors" rather than the ghosts (Greydanus et al. 2018). Attention-augmented agents, by contrast, scan the paths ahead for ghosts (Mott et al. 2019). Our agents are fully trained APPO agents, so both comparisons are loose.
- **Reconciling with Goodfire.** Goodfire's labels are free-text visual descriptions, which can express such conjunctions. Our result is a statement about a ground-truth *variable* basis. The conjunction analysis is post hoc and should be tested with a preregistered conjunctive dictionary.

Can single units steer behaviour?

- **Causally yes; specifically, only partly.**
  - *What holds.* Patching a single conv3 unit reliably changes behaviour on held-out games, beyond all 63 other channels at the same position, and the same selection rule finds such a unit in both other agents (H3).
  - *What the post-hoc checks remove.* The approach drop is not local to the steered cell. The selected unit is always the strongest perturbation at its cell, so the dose-rule null does not control for strength. The agent clearly leaves the steered region only in agent 3333, partly in 2222, and not in 1111.
  - *Reading.* The conv3 result is best described as "the strongest single-cell patch disrupts goal-directed navigation". It is not "a unit that encodes a place to avoid". A strength-matched null (doses chosen to match the candidate's open-loop KL, on fresh seeds) is the experiment that could separate the two; we did not run it.
- **Steering is destructive.** Every frozen candidate in every agent raises deaths per 1,000 steps and lowers the return (off-target tables). Under the conv3 patch, agents 1111 and 3333 also dither more; agent 2222 does not. Single-unit steering here is a blunt instrument, not a precise control knob.
- **Mechanism vs outcome.** The fc ablation shows a gap between mechanism and outcome. The predicted local behaviour ("go toward the edible ghost in direction d", a secondary metric; post hoc, no null ablation matches it, and it is concentrated in that context) changes strongly in agents 1111 (0.505 → 0.315 (Δ -0.190 [-0.209, -0.171])) and 3333 (0.476 → 0.307 (Δ -0.169 [-0.188, -0.149])), but barely in agent 2222 (0.549 → 0.526 (Δ -0.023 [-0.045, 0.001])). The global outcome (ghosts eaten per pill) changes little in agent 1111, falls in agent 2222, and *rises* in agent 3333. Removing one direction's vote appears to reroute the agent rather than simply weaken ghost hunting.
- **Why the fc layer is constrained.** Because the policy head is linear and has no recurrence, clamping an fc unit can only swap its vote for a fixed action bias, and ablation can only remove the vote. State-dependent steering must enter upstream, where activations pass through the fc ReLUs.

Methodological lessons.

1. Bit-exact fidelity tests are cheap and necessary, and must cover the inference device. Library defaults changed under us (gymnasium's grayscale). Our CPU-only gate missed a 4.5e-02 logit difference from cuDNN TF32 on the GPU. The A/A arm shows that it changed no sampled action in 50 of 50 paired games (agent 1111).
2. Near-deterministic policies need injected stochasticity (sticky actions) before games can serve as independent samples.
3. Effects selected on a noisy outcome shrink on held-out data (winner's curse). fc 453's pilot drop in ghosts per pill (0.392 → 0.140 (Δ -0.252 [-0.331, -0.171])) fell to Δ -0.066 [-0.192, 0.101] on held-out games. Event-level effects did not shrink: approach went from 0.430 → 0.375 (Δ -0.056 [-0.068, -0.044]) to Δ -0.063 [-0.077, -0.050]. Confirmatory effect sizes must come from held-out games.
4. A preregistered difference test needs a minimum effect size. Our H1 criterion was "passed" by a difference of 0.6% [0.2%, 0.6%].
5. A behavioural metric that scores one target can pass a null test without being specific to that target. Score the same metric on control targets (here all 49 cells), and match the null on perturbation strength, not only on the dose rule.

6. Limitations

- **Planned but not done.**
  - An input-pixel counterfactual for the top candidates (repainting ghosts as frightened in the observation) to separate ghost appearance from correlated game state.
  - A per-ghost "visible in the observation" flag (Atari sprite flicker can hide a ghost that RAM reports).
  - Training dynamics over the 36 milestone checkpoints of agent 1111.
  - Replication on agents trained with other algorithms (Stable-Baselines3 DQN/PPO).
  - The OCAtari cross-check of RAM decoding, replaced by pixel-level validation. OCAtari releases ≥ 1.1.6 pin numpy < 2, which is incompatible with this project's numpy ≥ 2.5, and the 1.1.5 source distribution failed to build in our environment.
- **Inference device.** All rollouts ran on a GPU with PyTorch's default cuDNN TF32 convolutions, which the CPU fidelity gate does not cover: logits differ from Sample Factory by up to 4.5e-02. The agent we analyse is therefore a very close numerical neighbour of the trained one, not a bit-exact copy. The A/A arm shows that the seed pairing is intact.
- `ram[116]`. Its upper bits change when ghosts are eaten (set on 18.5% of frightened steps), and the frozen `fright_timer` feature used the raw byte. Masking it (exploratory) raises one feature's split-half R^2 by at most 0.047 and makes no unit "clean" (0 units).
- **Nulls and clusters.** Each null unit is estimated from 20 games against the candidate's 50, which widens the null distribution and so raises the bar for the specificity criterion. The conv3 null matched the dose rule, not perturbation strength: the selected patch is always the strongest at its cell (Section 4.3). The start-state-cluster bootstrap has only 4 clusters. The intraclass correlation of the paired Δ within clusters is 0.000–0.125. With a design-effect correction, every preregistered pass stays below 0.05; the closest is fc379_ablate in appo_2222 (cluster-adjusted Holm p 0.0112).
- **Metric construct validity.** The preregistered approach metric scores one target node. Post hoc it falls for almost every node, so a pass on it does not show spatial specificity.
- **Post-hoc analyses.** The specificity, context and dependence checks were added by an audit after every confirmatory result was known. They use no new games, but they are exploratory.
- **Occlusion.** Maps average a unit's 50 top states. For fc 453 these mostly repeat one early-game moment, and occlusion cannot separate Ms. Pac-Man from the maze patch she occupies.
- **Feature dictionary.** The one-job criterion depends on a finite 40-feature dictionary of *single* variables, so conjunctive codes are counted as "not clean" by construction. The conjunction analysis is post hoc.
- **Sticky actions.** These change the agent's task (it was trained without them) and lower its score. All results describe the agent under p = 0.25.
- **Dependence between games.** Games that share a start state overlap for their first few hundred steps. We report a start-state-cluster bootstrap as a robustness check.
- **Scope of interventions.** Steering was tested at fixed doses (discovery p99) and only for the frozen intervention types; dose-response is exploratory.
- **Agents.** Three public agents of one architecture and algorithm. Generalisation to other algorithms (DQN, PPO with IMPALA-ResNet) is untested. The untrained control uses the same initialisation seed for all three agents.

7. Conclusion

Clean? In three reward-trained Ms. Pac-Man agents, individual neurons are not "already clean" with respect to exact single game variables. They are sparse and, by exploratory evidence, conjunctive, place-like codes.

Steerable? Causally yes, specifically only in part.

- *conv3.* A single conv3 patch passes the preregistered steering test beyond all 63 dose-matched null channels, and its selection rule replicated in both other agents. Post-hoc checks show what it does: it is the strongest perturbation at its cell and disrupts navigation across the maze. It makes the agent clearly avoid the steered region in one of three agents (3333), and partly in a second (2222).
- *fc.* Zero-ablating a single fc unit reduces a predicted piece of ghost-chasing (moving up toward an edible ghost). That reduction is specific to its context and to the unit, but it does not significantly change ghosts eaten.
- *Cost.* These effects come with substantial collateral damage.

Toolkit. The fidelity-checked models, RAM ground truth, preregistration and microscope make these agents a convenient testbed for future work: conjunctive dictionaries, sparse autoencoders, and other algorithms.

Appendix A. Fidelity details

See `reports/01_fidelity.md`.

Appendix B. Feature list

See `src/pacneuron/features.py` and `reports/02_ram_ground_truth.md`.

Appendix C. Preregistration and deviations

See `reports/preregistration.md`, `reports/preregistration_addendum.md` and `reports/00_summary.md`. The frozen files are left unchanged; corrections to them are listed under "Errata and undeclared deviations" in `reports/00_summary.md`.

AI statement

This codebase and paper were written by Claude Code (Anthropic's Claude), following an exact specification from a human author who directed the research. Human cleanup and rewriting are in progress.